

Shubhangi Mishra

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SUMMARY

Senior Software Engineer with 6+ years of experience building Java-based backend, distributed, and full-stack applications across financial services and retail. Specializes in Java, Spring Boot, microservices, Kafka, AWS, Kubernetes, and PostgreSQL, delivering high-volume transaction processing, event-driven services, customer-facing APIs, and web applications using React and Angular. Hands-on experience with Generative AI, RAG, LLM workflows, system design, distributed systems, performance engineering, secure APIs, automated testing, CI/CD, observability, and production reliability.

SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript

Backend & Frontend: Spring Boot, Spring MVC, JPA, Hibernate, REST APIs, Microservices, gRPC, React, Redux, Angular

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, CI/CD, Linux

Data & Messaging: PostgreSQL, MySQL, Oracle, MongoDB, Redis, Kafka, RabbitMQ, Elasticsearch

AI & GenAI: RAG, LLMs, Prompt Engineering, Embeddings, Vector Search

Security & Testing: Spring Security, OAuth2, JWT, RBAC, JUnit, Mockito, Integration Testing

Engineering & Observability: System Design, Distributed Systems, Concurrency, API Design, Scalability, Performance Optimization, Prometheus, Grafana, ELK, Distributed Tracing

PROFESSIONAL EXPERIENCE

Capital One

May 2024 – Present | New York, Hybrid

Senior Software Development Engineer

- Architected Java/Spring Boot microservices for credit-card account servicing and digital customer workflows, applying validation, authorization, idempotency, and recovery patterns across distributed services processing 600K+ requests daily.
- Optimized REST APIs backed by PostgreSQL and Redis through SQL tuning, indexing, connection-pool optimization, and caching, reducing median transaction and balance lookup latency from 118 ms to 34 ms.
- Engineered Kafka-based event processing for card and account workflows, handling 95K+ events per hour for transaction monitoring, customer notifications, and downstream services with retry, failure recovery, and offset management.
- Developed transaction-insight services correlating merchant, customer, account, and transaction signals across 1.2M+ monthly records, enabling near-real-time identification of unusual activity, duplicate transactions, recurring charges, and other account events.
- Integrated gRPC and RabbitMQ across asynchronous account-servicing workflows, defining service contracts, timeout and retry policies, dead-letter handling, and idempotent consumers across 15+ dependent services.
- Modernized legacy account-processing and reconciliation workloads into Kubernetes-based services, introducing parallel processing, controlled retries, and fault isolation to reduce overnight processing time from 3.6 hours to 1.4 hours.
- Contributed to a GenAI-powered RAG workflow for internal payment and account-operations support, improving retrieval and prompt-processing flows across 30K+ operational knowledge records and saving approximately 5 hours of investigation effort per week.
- Improved production reliability and observability across 20+ services using Prometheus, Grafana, ELK, distributed tracing, and Kafka consumer-lag monitoring, giving engineering teams better visibility into failures and recurring processing issues.

JP Morgan Chase & Co.

Jun 2023 – Apr 2024 | New York

Software Engineer

- Built Java/Spring Boot services for payment initiation, transaction validation, settlement updates, and exception handling, incorporating idempotency, audit logging, validation, and recovery into banking workflows.
- Refactored account-servicing capabilities into 20+ REST APIs and built React/TypeScript applications with Redux for customer operations, reducing average request processing time from 8 seconds to under 3 seconds through API and database optimization.
- Hardened customer onboarding and KYC workflows with Spring Security, RBAC, OAuth2/JWT, audit trails, and validation controls, lowering onboarding exceptions by 35% across 100K+ customer records.
- Contributed to system design for 25+ banking services, defining API contracts, service dependencies, failure-handling paths, and deployment requirements, helping reduce environment provisioning time from 2 days to under 4 hours.
- Automated reconciliation across 12+ financial systems, validating transaction and settlement records and routing exceptions through repeatable backend workflows instead of spreadsheet-based checks.
- Expanded automated testing for critical banking services using JUnit, Mockito, and integration testing, supporting UAT, production releases, and troubleshooting of recurring transaction-processing issues throughout the SDLC.

Accenture

Apr 2018 – Dec 2021 | India

Software Engineer

- Constructed Java/Spring Boot microservices for retail clients across order management, cart, checkout, pricing, and customer workflows, supporting 10+ production services.
- Developed reusable Angular components for product browsing, cart management, checkout, and order tracking, integrating 30+ REST APIs across customer-facing retail journeys.
- Engineered Kafka event pipelines for order confirmation, payment processing, inventory synchronization, and fulfillment, processing 35K+ events per minute across distributed retail services.
- Improved MySQL and PostgreSQL data layers through schema design, query tuning, indexing, and partitioning across customer, product, inventory, and order data, reducing critical query time from 17 seconds to under 5 seconds.
- Added Redis caching and Elasticsearch to catalog and search workloads, reducing product-search latency from 900 ms to 310 ms during testing.
- Improved CI/CD delivery with Jenkins, Docker, AWS, and Maven, supporting 15+ monthly releases across development, QA, and production and cutting deployment time from 18 minutes to 8 minutes.
- Built Python automation for inventory reconciliation and data validation, processing 100K+ SKU records daily and eliminating more than 2 hours of manual verification from recurring operational work.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner

EDUCATION

Master of Science in Computer Science

Jan 2022 – Jun 2023 | New York, USA

University at Buffalo